

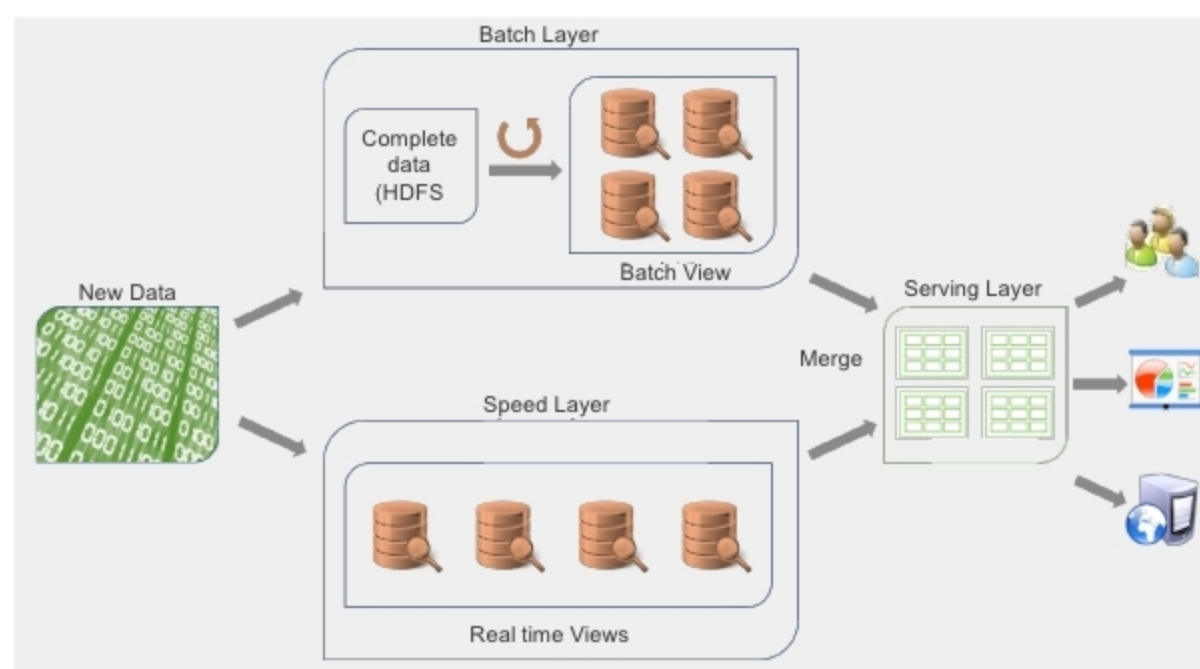


Figure 2: Lambda architecture

Lambda Architecture

It could be challenging to design and build a Big Data solution that is fault-tolerant, scalable and caters to diverse use cases and workloads with different latency requirements. The Lambda Architecture is a generic framework that could be leveraged to design robust and reliable Big Data applications. It proposes a three-layered architecture consisting of the batch layer, speed layer and serving layer.

- All new data entering the system is sent to both the batch layer and speed layer.
- **Batch layer:** This caters to two primary needs. One is to maintain a master data set and the other is to pre-compute batch views.
- **Speed layer:** This deals with only recent data and caters to all low latency requirements.
- **Serving layer:** This merges data from both the batch and speed layers, and caters to downstream data needs.

To complement the implementation and data architecture frameworks, enterprises must also take care of certain considerations when working with Big Data.

Key considerations with Big Data

To leverage Hadoop based systems as a core data layer, it is imperative that enterprises have proper governance and controls in place. Certain

important aspects that need to be considered when dealing with Big Data include the following.

- **Data ingestion:** This helps to simplify the on-boarding of new data assets and boosts the ability to develop new use cases. A managed process should be in place to decide what data gets ingested from where, at what time and where it gets stored within the data layer.
- **Metadata management:** This is the ability to make a data catalogue (containing all the metadata captured during the data acquisition and validation) available to all the stakeholders.
- **Security:** This is the ability to authorise users and control access to data assets.
- **Compliance:** This monitors PCI, PII, PMI and statutory compliance of how data assets are used.
- **Technology changes:** Hadoop based ecosystems, which are primarily used with Big Data, are robust, complex and constantly evolving. In addition to new components being constantly added to the Hadoop ecosystem, each component by itself is being enhanced. To keep up with the latest and best developments in the Hadoop open source world and embrace them could be a challenging task.
- **Data archival policy:** It is very important to have a comprehensive

data archival and purging policy in place. Data should be segregated into hot, warm and cold tiers and should be organised accordingly.

- **Audit and logging:** These functions maintain the details of the all-important activities taking place in the Big Data ecosystem.

The current buzz words in the tech world include the cloud, artificial intelligence and machine learning. In the context of Big Data, these technologies are not only very relevant but also make up an extremely desirable combination.

The cloud – a catalyst for Big Data initiatives

One of the main reasons enterprises experience delays in Hadoop initiatives taking off is the effort and skills needed to procure and set up Hadoop clusters on-premise or in their data centres. In addition, making huge investments for new business initiatives may not always be a viable option. Here is where enterprises can look to the cloud as a testbed for their new ideas. Cloud providers like Microsoft offer managed Hadoop services like HDInsight and Azure Databricks, which can be leveraged to jumpstart Big Data initiatives.

Some of the reasons why Hadoop on the cloud is a win-win situation are listed below.

- **Increased agility:** Setting up Hadoop on-premise can take up to a few days and would require skilled Hadoop resources. This set-up time is drastically reduced to just a few minutes on the cloud without the need of any expert Hadoop resource. All it takes is a few clicks from a browser and a few minutes to set up a Hadoop cluster on the cloud.
- **Optimised costs:** Enterprises can start with a small cluster that has a few nodes and add more nodes as per their needs. They also have an option to shut down the cluster during non-business hours to further optimise expenditure
- **Reduced cycle time:** Many enterprises are leveraging the virtually infinite